



Problem & Objective

Problem

Wildlife monitoring in the Arctic fails for two reasons.

1. Always-on computing architectures drain batteries within days in sub-zero environments, where battery chemistry itself degrades. LiFePO₄ — the standard choice for remote deployments — cannot accept a charge below freezing and suffers severe voltage sag under load, making it hard to estimate battery life.
2. Single-stage motion detection systems are defeated by the Arctic environment itself: wind-blown snow, sun-warmed ice surfaces, and rapidly shifting shadows trigger continuous false positives, wasting what little power remains on recording empty footage and filling storage with useless data.

Objective

Design and deploy a fully autonomous, self-contained 360° wildlife observation station capable of surviving 90 days unattended in sub-zero Arctic conditions. The system must capture 11K RAW imagery on a fixed interval and record motion-triggered video only upon dual-verified wildlife detection — aggressively rejecting false positives without human intervention.

Dual-Node Architecture

The core insight of Polar Eyes is asymmetric computing — permanently separating the always-on monitoring function from the high-power media acquisition function, making a 90-day battery budget achievable.

Sentry Node — Arduino ItsyBitsy 32u4

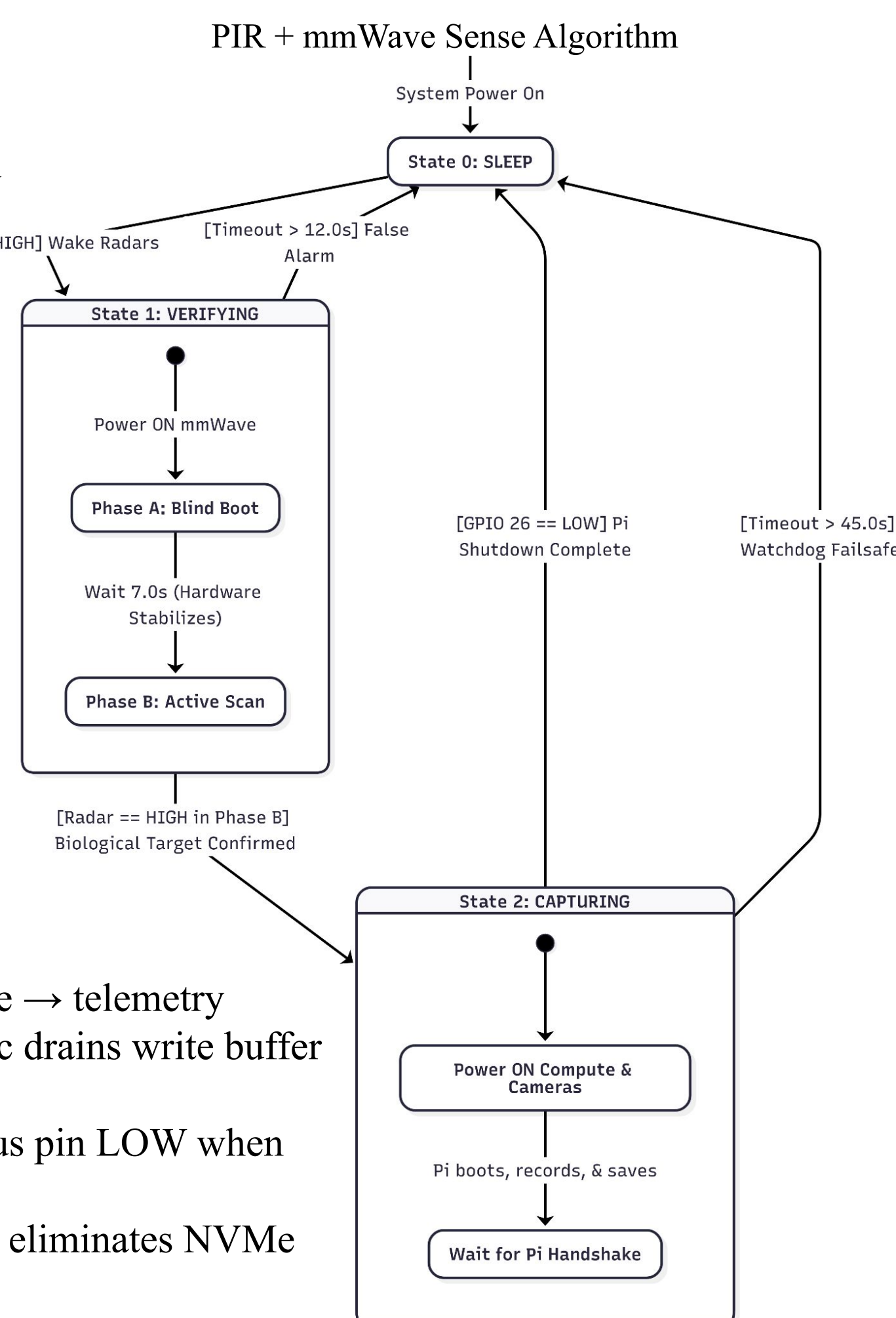
The always-on brain, drawing <10 mW. Continuously polls 3× AM312 PIR sensors (D11/D12/D13) at only ~50 µA total — essentially free from a power budget perspective. On a PIR rising edge, gates 5V via MOSFET to the 3× SEN0395 mmWave array (~56 mA active). Enforces a 7-second "blind boot" via non-blocking millis() to discard floating-pin noise during radar startup, followed by a 5-second active scan to confirm biological motion. Also maintains the 10-minute timelapse clock and polls the INA260 over I²C for real-time power telemetry.

Worker Node — Raspberry Pi 5

Hard-off at 0 W until the Sentry flips the opto-relay. On boot, systemd auto-starts polar_listener.py, which reads BCM 27/24 to determine photo vs. video mode and executes the Insta360 X5 SDK capture sequence over the powered USB 3.0 hub.

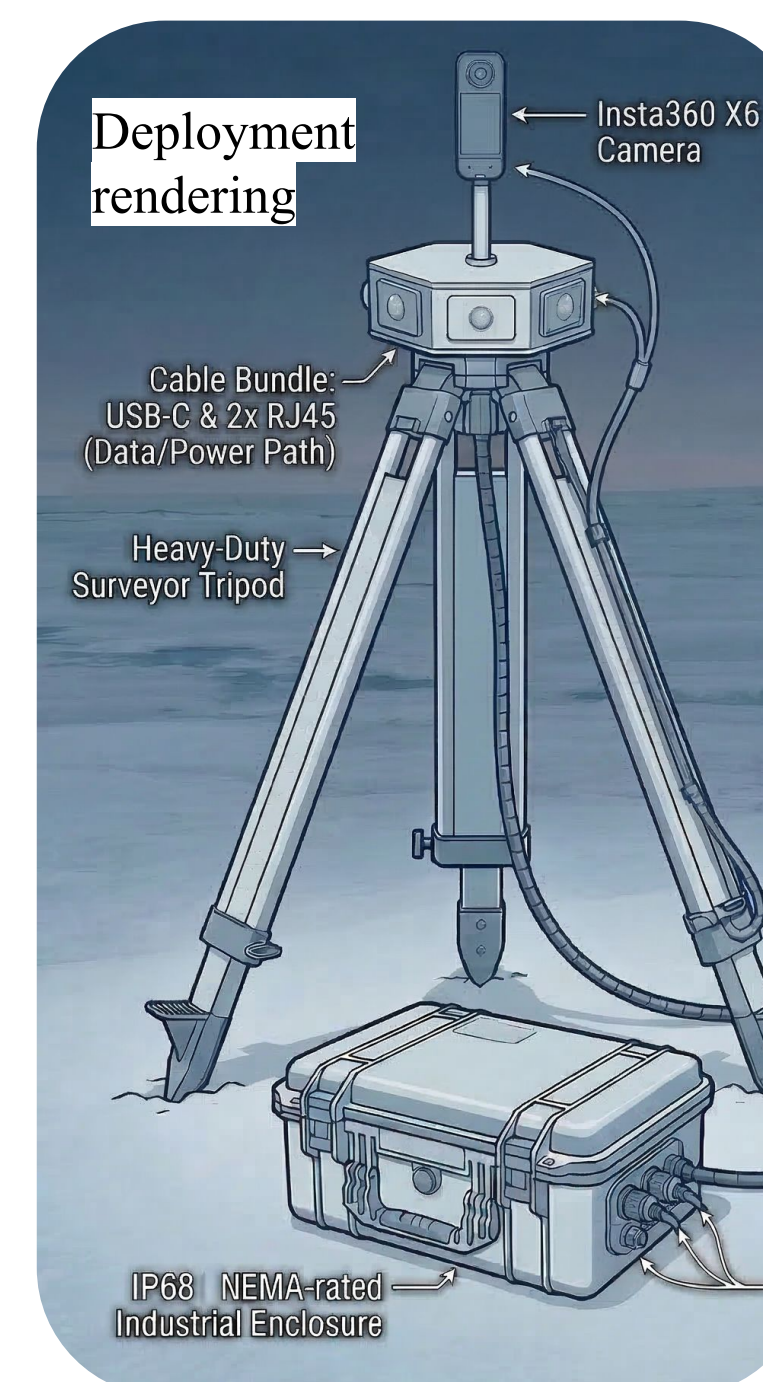
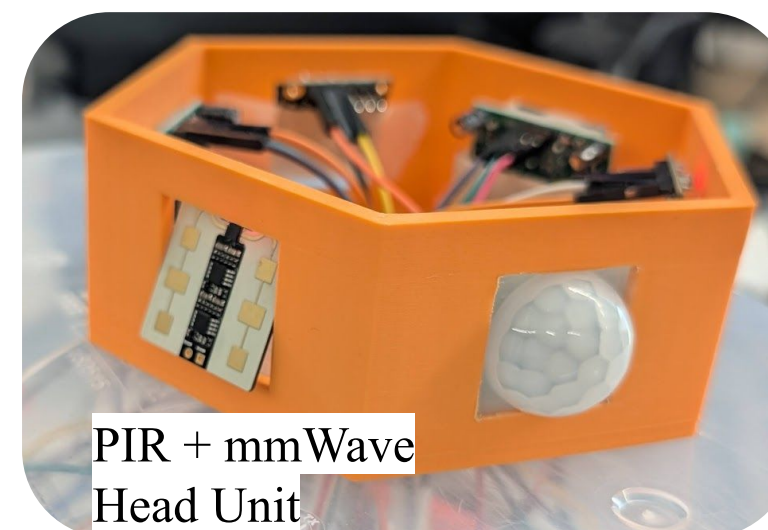
Graceful Shutdown Handshake

- Camera SDK signals capture complete → telemetry logger flushes CSV → filesystem sync drains write buffer → Pi issues shutdown -h now
- Pi 5 PMIC pulls hardware power-status pin LOW when OS fully halted
- Power is never removed mid-write — eliminates NVMe corruption risk



System Integration

Event Type	Frequency	File Size (11K RAW)	Daily Accumulation	90-Day Total
Timelapse Photo	6 per hour (144/day)	~30 MB (.insp)	4.3 GB / day	388 GB
Motion Video (60s)	Est. 1 per day	~800 MB (.insv)	0.8 GB / day	72 GB
System Telemetry	145 logs/day	< 1 KB (.csv)	< 1 MB / day	< 10 MB
TOTAL			~5.1 GB / day	~460 GB



Dual Sense: PIR + mmWave

To solve the Arctic false-positive problem without breaking the energy budget, Polar Eyes uses a cascaded, two-stage verification architecture.

PIR (Power: Low | Accuracy: Low)

- Responds to any infrared delta (sun-warmed ice, wind-blown snow, tidal shifts, shadows/clouds).
- A PIR-only system would trigger constantly, wasting storage and killing the battery with false recordings.

mmWave (Power: High | Accuracy: High)

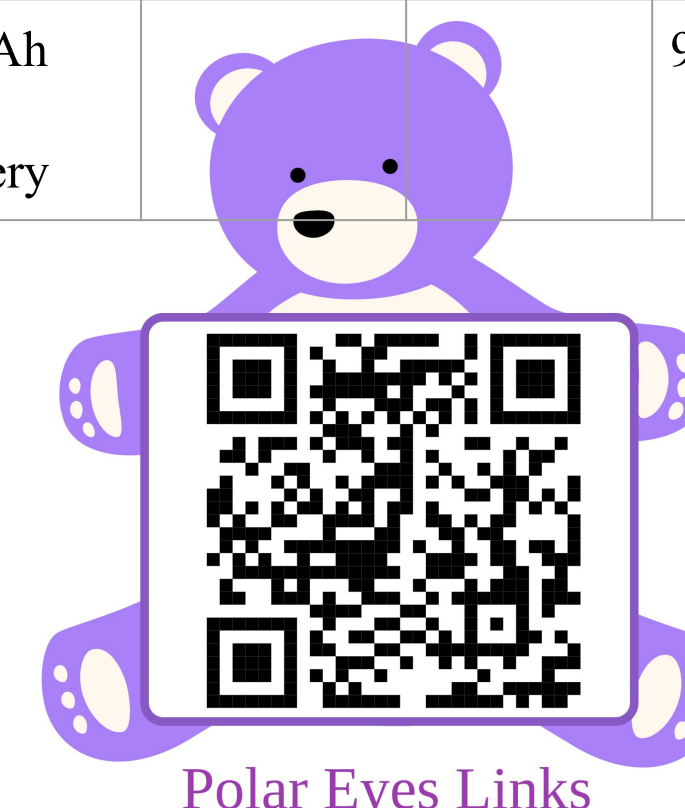
- Perfectly detects actual physical movement via Doppler shift, ignoring thermal noise.
- The Power Cost: Each SEN0395 draws ~56 mA (280 mW). An always-on 3-radar array would consume 1,814 Wh over 90 days—vastly exceeding the battery budget.

Dual Sense (Power: Low | Accuracy: High)

- Stage 1 (Always-On): PIR provides near zero-cost continuous coverage.
- Stage 2 (Gated Verification): mmWave activates only when PIR provides "probable cause," consuming power only during a brief, multi-second verification window.

Sensor	Current Draw	Always-On 90-Day Cost
AM312 PIR (×3)	~50 µA total	~0.11 Wh
SEN0395 mmWave (×3)	~56 mA each	~1,814 Wh (if always-on)
mmWave (gated)	~56 mA × 12s/event	Negligible

Component	State	Current	Daily Energy
Sentry (Arduino)	Always-on	~8 mA @ 5V	~0.96 Wh
PIR Array (×3)	Always-on	~50 µA	~0.014 Wh
Raspberry Pi 5	~2 min/cycle	~900 mA @ 5V	~2.16 Wh
Insta360 X5	Capture only	~400 mA	~0.32 Wh
mmWave Array (×3)	Gated, ~12s/event	~168 mA	Negligible
Iridium Modem	Pi-on only	Peak 1.5A	~0.25 Wh
Total Estimated			~3.7 Wh/day
51 Wh Battery			~13 day alpha
300 Ah Beta Battery			90+ days ✓



Results & Autonomous Operation

Validated Performance

- End-to-end cascaded architecture (PIR → mmWave → Worker Node → Camera) successfully demonstrated.
- Stage-2 mmWave verification successfully implemented; actively rejects non-biological thermal noise before waking the Raspberry Pi.
- Video GPIO holds HIGH for full event duration; re-triggers suppressed by firmware state machine
- Timelapse and motion-triggered media saved locally to NVMe storage with perfectly synchronized I²C telemetry appended per capture.
- While a full-system physical freezer test was descoped due to time constraints, all critical power-path components (Sodium-Ion cells, Pololu MOSFETs) are datasheet-validated for continuous operation down to -20°C.

Iridium Data Packet Design

- Network caps payloads at 340 bytes — packet engineered to stay under 30 bytes
- GPS coordinates packed as 4-byte fixed-point integers vs. ~22-byte ASCII floats → 8 bytes vs. 22
- Temperature, pressure, voltage quantized to 2-byte unsigned integers with known scale factors
- 1-byte status bitmask encodes 8 boolean system flags (mmWave armed, camera active, storage healthy, etc.) in a single byte

Fault Tolerance

- Every Insta360 SDK call wrapped in a Bash timeout guillotine
- SDK has a known hang-at-99% bug on network socket close during video stop
- Wrapper enforces hard 15-second ceiling, logs timeout, continues shutdown regardless
- System always returns to idle — no manual reboot ever required in the field

Telemetry Sample (.csv)

Timestamp,Event,Lat,Lon,Press(hPa),Temp(C),Batt(V),Draw(mA)
2026-04-14 06:00:00,INTERVAL_10M,42.3601,-71.0589,1012.5,-14.2,12.5,450
2026-04-14 06:10:00,INTERVAL_10M,42.3601,-71.0589,1012.4,-14.5,12.4,440
2026-04-14 06:12:34,MOTION_VERIFIED,42.3601,-71.0589,1012.4,-14.5,12.3,1250

Power & Survivability

Battery Selection — Sodium-Ion

- Powered by a 12V Sodium-Ion battery — deliberate departure from LiFePO₄ standard
- Sodium-ion maintains discharge curve well below freezing
- LiFePO₄ not only loses capacity in cold but refuses to accept a charge entirely — fatal for any solar-rechargeable arctic deployment

Split Switching Topology

- Pi 5 powered exclusively through an opto-relay — complete electrical isolation, zero voltage drop
- All other high-draw peripherals (mmWave array, USB hub) switched by Pololu solid-state MOSFETs
- MOSFETs draw 0 mA holding current vs. ~80 mA vampire drain of mechanical Songle relays replaced
- 80 mA × 90 days = 172.8 Ah wasted doing nothing — eliminated entirely

Iridium Rail Isolation

- Iridium modem powered by dedicated independent buck converter (Buck B)
- During satellite transmission, modem pulls 1.5 A spikes at 5V
- Without isolation, spikes sag Pi's 5.1V compute rail mid-write — known cause of NVMe corruption
- Buck B ensures Pi compute rail stays clean regardless of modem activity

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